# Capstone Project Report
# SentinelTrap Enterprise

A Unified Cybersecurity Operations & Threat Intelligence Dashboard

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## 1. Introduction

In modern Security Operations Centers (SOCs), analysts are frequently burdened by "tool fatigue." Security assessments require the use of disparate, disconnected command-line utilities—such as Nmap for network discovery, Burp Suite for web fuzzing, and custom scripts for log parsing. This fragmentation creates operational inefficiencies, increases the Mean Time to Detect (MTTD), and increases the likelihood of human error during critical incident response scenarios.

To solve this challenge, this project introduces **SentinelTrap Enterprise**: a centralized, web-based Security Dashboard. By consolidating offensive assessment tools and defensive monitoring capabilities into a single, authenticated web interface, analysts can execute comprehensive security assessments and monitor live adversarial threats from a single pane of glass.

## 2. Abstract

SentinelTrap Enterprise is a full-stack cybersecurity application developed using Python and the Flask framework. The platform utilizes a modular, microservice-like architecture to host four distinct security modules:

- **Deception Technology:** A live SSH Honeypot (built with Paramiko) that mimics a vulnerable host, logging unauthorized brute-force attempts and credentials into a local SQLite database.
- **Web Application Security:** A dynamic scanner that utilizes BeautifulSoup to crawl target endpoints and inject XSS payloads to identify input sanitization flaws.
- **Network Reconnaissance:** A high-speed TCP port scanner leveraging Python's native socket library to map exposed services.
- **Incident Triage:** A Pandas-powered log analyzer that ingests web server logs to detect brute-force patterns (HTTP 401 spikes).

To ensure enterprise compliance and data integrity, the dashboard enforces strict Role-Based Access Control (RBAC) via session-based authentication, ensuring that sensitive reconnaissance capabilities are restricted to authorized personnel only.

## 3. Tools Used

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|---|---|
| **Python 3** | Core programming language driving backend logic and data processing. |
| **Flask Framework** | REST API routing, session management, and web templating. |
| **Paramiko** | SSH protocol emulation for the deception honeypot. |
| **SQLite3** | Lightweight relational database for capturing attack telemetry. |
| **BeautifulSoup & Requests** | DOM parsing, HTTP requests, and XSS payload injection. |
| **Pandas** | High-performance data manipulation and threat detection in server logs. |
| **HTML5 / CSS3 / JavaScript** | Responsive, asynchronous, dark-mode UI (Fetch API). |

## 4. Steps Involved in Building the Project

**1. Architecture & Authentication:** Configured the Flask application with a custom session-based login portal, ensuring all internal APIs are protected by authentication guards.

**2. Honeypot Development:** Built a custom SSH server that mimics a vulnerable Linux environment. When brute-forced, it captures the attacker's IP, username, and password.

**3. Web Fuzzing Engine:** Created a crawler that extracts HTML <form> tags and actively tests them by injecting <script> payloads, analyzing the HTTP response to confirm reflected XSS.

**4. Network Reconnaissance:** Engineered a high-speed TCP socket scanner that tests common ports (21, 22, 80, 443, 3306) to identify open services on a target IP.

**5. Log Analysis Pipeline:** Programmed a script that ingests raw web server logs, groups failed login attempts by IP address, and flags IPs exceeding the brute-force threshold.

**6. Dashboard Integration:** Connected all backend APIs to a unified frontend dashboard using JavaScript Fetch API, allowing the analyst to run all four tools seamlessly without page refresh.

## 5. Application Screenshots

The following screenshots demonstrate the live, authenticated interface of SentinelTrap Enterprise running on the local development environment (127.0.0.1:5000).

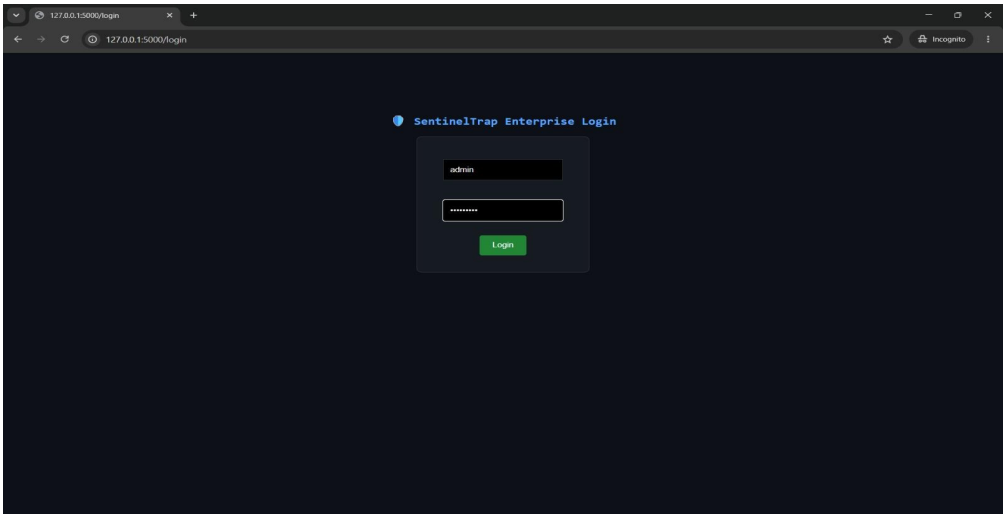



*Figure 1: Secure Login Portal — Session-based authentication with RBAC enforcement.*

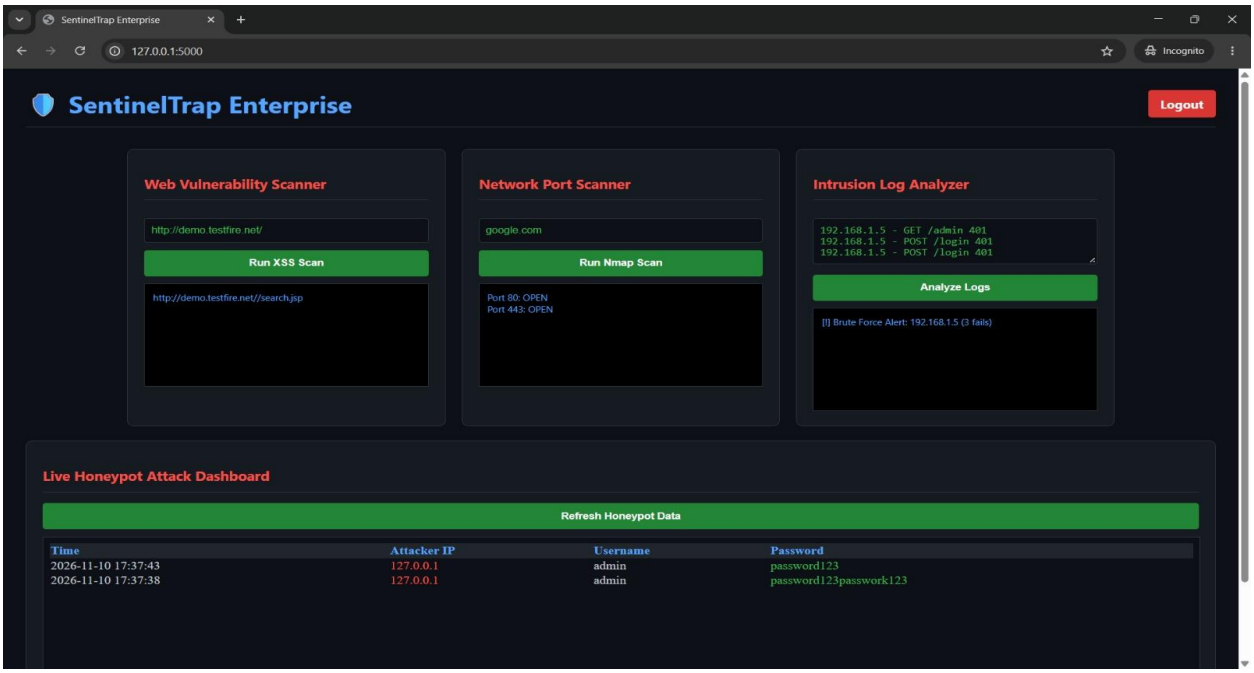



*Figure 2: Main Dashboard — Web Vulnerability Scanner, Network Port Scanner, Intrusion Log Analyzer, and Live Honeypot Attack Dashboard in a unified dark-mode interface.*

## 6. Conclusion

SentinelTrap Enterprise successfully bridges the gap between offensive and defensive cybersecurity operations. By consolidating diverse security modules into a single, password-protected platform, the project demonstrates how automation can drastically reduce analyst workload and accelerate incident response.

The project proves advanced competency in full-stack Python development, API architecture, database management, and the practical application of the OWASP Top 10. Furthermore, it highlights the value of deception technology in gathering actionable threat intelligence against automated botnets.

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*This report documents the design, implementation, and operational capabilities of SentinelTrap Enterprise as submitted for the Elevate Labs Internship Capstone Project.*